Learning to Manipulate a New Object from Action-Free Videos

Group 9
McGill University
ECSE 552 — Project proposal

Background

Application. We teach a robot arm that manipulates one object to handle new ones from videos alone: warehouse and home robots meet new items constantly, and recording paired images and motor commands for each requires operating the robot. We study this in OCBench simulation (UR5e arm, Robotiq gripper) [1].

Problem and need. Visuomotor policies imitate images paired with motor commands, and acquiring such pairs across varying configurations is costly and time-consuming: 50 people took a year to collect 76k teleoperated demonstrations [2]. With limited training coverage, a policy may produce an erroneous action when it encounters an unfamiliar situation. Errors can compound during execution, moving the robot further from situations represented in its training data [3]. We investigate whether action-free videos can help pixel-to-action policies manipulate new objects without collecting additional action labels for those objects.

Constraints.

ID	Requirement	Definition	Target
R1	Supervision	Target data are RGB frames only	Zero target labels
R2	Label budget	Labeled cube demonstrations, identical for every method	100 demos; 300 if cube-only ACT is below 80% in the source pilot
R3	Inputs	No simulator state in any learned model	Images only
R4	Target	Success gain over the cube-only policy on each target, as a fraction of the oracle's gain	$\geq 50\%$ of gap
R5	Retention	Drop in cube success after adding target data	≤ 10 points
R6	Real time	Policy inference per call, completed within one 25-action chunk at 50 Hz, so it can run while the previous chunk executes (IDM offline)	< 0.5 s

Aims/Goals

Goals. Meet R4 and R5 on each target, satisfy R1–R3 by design, and measure R6.

Scope. We study an alternative data-gathering method in simulation, evaluating three target objects and an optional fourth; real robots are unavailable. We use OCBench's visual single-block task with its default $6 \times 6 \times 6$ cm red cube as source. Targets are (A) a blue cube of identical size (appearance), (B) a $9 \times 6 \times 6$ cm red block (geometry, same grasp width), (C) a $9 \times 7 \times 6$ cm red block (geometry and grasp width), and optionally (D) a blue version of C (combined). Expert grasps are adapted; robots, cameras, lighting, and tasks stay fixed. Human videos and large pretrained models are out of scope.

Proposed solution. World models and video generation models could provide demonstrations without action labels [4]. In this project, we use simulator-generated videos of new objects to evaluate learning from action-free demonstrations in a controlled setting. We feed consecutive frames into an inverse dynamics model (IDM), trained on few labeled cube demonstrations [5], and infer actions for imitation learning [6], [7]. Formally, given cube transitions $\mathcal{D}_A = \{(o_t, o_{t+1}, a_t)\}$ and unlabeled target videos $\mathcal{D}_B = \{o_t\}$, we seek an image-only policy $\pi(a \mid o)$ satisfying R1–R6. We adapt the IDM to each target before labeling its videos, using frame differences for motion cues and DANN [8] to encourage domain-invariant features while still predicting cube actions. One ACT policy per target then trains on cube demonstrations plus pseudo-labeled target videos. Adaptation should help most on A, where only appearance changes.

Methodology

Design and implementation. Observations are 224×224 RGB images from the front, side, and wrist cameras; actions are seven normalized joint and gripper increments at 50 Hz. The IDM uses a convolutional encoder shared across cameras to embed consecutive frames and their difference, followed by an MLP trained with action mean squared error. Following DANN [8], a domain discriminator is trained with binary cross-entropy through a gradient reversal layer, whose reversed gradient is scaled by a weight λ that ramps up during training. The IDM retains its supervised source action-prediction loss; $\lambda = 0$ gives the unadapted IDM. The policy is an image-only adaptation of ACT [3], excluding joint-position inputs and predicting chunks of 25 action increments. It is trained as a conditional VAE with L1 reconstruction and KL-divergence losses. Chunks are executed open loop without temporal ensembling. IDM and policy encoder weights are separate. All policy comparisons use the same ACT architecture and training budget in PyTorch.

Validation, testing, and evaluation. Demonstrations are split by episode into 80% training, 10% validation, and 10% test, with identical splits across methods. Target actions never select models: λ is prescheduled, IDM checkpoints follow source validation error, and policy training steps are fixed. Across three seeds, we report per-component IDM error on held-out test transitions, policy inference latency, and target success and cube retention over 100 separate evaluation rollouts per seed, method, and object, with identical initial states across methods. We compare ACT policies using (a) cube-only data, (b) unadapted IDM labels, (c) DANN-adapted IDM labels, (d) true target actions, and (e) cube-only augmentation; optionally, (f) adds policy-encoder alignment to (b). Target actions otherwise serve only post-training evaluation. R4 is inconclusive if (d)'s success does not exceed (a)'s.

Alternatives. Methods (b)–(d) also use cube demonstrations; (b) and (c) share augmentation settings. Comparing (b)/(c) isolates IDM adaptation; (e)/(c) tests gains over strong source-only augmentation; (a)/(d) anchor R4. Method (e) uses hue, brightness, contrast, and saturation jitter and small random crops. We expect it to be competitive on A; B and C test geometry and grasp-width changes these augmentations do not directly reproduce. We hypothesize that (c) improves pseudo-label accuracy and policy success over (b), while testing for negative transfer.

Feasibility and resources

Dataset, resources, and plan. We generate all data in OCBench with MuJoCo: 100 labeled cube demonstrations (300 if cube-only ACT scores below 80% in separate cube-validation rollouts; the budget and split ratios are fixed before any comparison) and 200 action-free videos per target, keeping true target actions only for evaluation and the oracle; no licensing or ethics constraints apply. Training uses one NVIDIA A10G GPU; the pilot estimates compute. A pilot by October 21 checks expert success, the cube-only policy's success on the cube and each target, and the IDM's source validation error. Core methods follow by November 1; evaluation by November 18.

Risk analysis

Pitfalls and mitigation. Adversarial alignment does not guarantee action alignment [9]; per-component errors test whether actions remain readable from arm motion. If experts fail, adjust objects or grasps; if cube-only ACT fails on the cube, use OCBench's shorter lite task. We report negative transfer if alignment hurts. If time runs short, drop target D and policy-alignment experiment (f); oracle (d) remains mandatory for R4.

Role of group members

All members collaborate and review the report.

Member	Role	Tasks
Brik Meza	Data and evaluation	Simulation, expert demos, datasets, splits, rollouts, metrics
Myriam Dardour	Adversarial adaptation	Discriminator, adversarial training, tuning, experiments
Zheng Ye Zhang	Inverse dynamics model	IDM architecture and training, pseudo-labeling, error analysis
Gonzalo Millán García	Baselines and alternatives	Policy training, baselines, alternative methods, comparisons

Generative-AI disclosure. ChatGPT (OpenAI) and Claude (Anthropic) assisted with literature search, bibliography formatting, document structure, and prose. We used them to survey recent work and to format the document; the team revised all text and verified every claim and citation against the sources.

References

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